IMPROVED UPPER AND LOWER BOUNDS ON THE DENSITY OF IDENTIFYING CODES IN THE TRUNCATED TRIHEXAGONAL (4.6.12) GRID

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Abstract. We prove that the minimum density of a radius-one identifying code in the infinite truncated trihexagonal grid lies between 1829/4263 and 45/104, improving Sampaio's interval $[3/7, 13/30]$ at both ends. A periodic code selects 135 of the 312 vertices in a fundamental region. For the lower bound, we prove that every ball of graph radius 22 has total surplus at least 2 under Sampaio's discharging rule. Averaging over centers gives the lower bound for every identifying code, including nonperiodic ones.

The upper value 45/104 is the exact minimum in two restricted classes. The first consists of codes invariant under the translation $(2, 1)$ in honeycomb coordinates. In the second, consecutive anchor columns are two or three columns apart. Each anchor carries one of the 26 column patterns of the periodic construction. The intervening one or two columns are subject only to the identifying-code conditions, with no periodicity assumption on the filling. The exact minimum on the unrestricted plane remains open.

1. The density problem

Let G be the graph formed by the vertices and edges of the infinite 4.6.12 tiling. A square, a hexagon and a dodecagon meet at each vertex. A radius-one identifying code is a set C for which the signatures $N[v] \cap C$ are nonempty and pairwise distinct over all vertices v, including those in C.

Write $B_R(w)$ for the closed ball of graph radius R about w. Following Sampaio, set

$$d(C) = \inf_{w \in V(G)} \limsup_{R \to \infty} \frac{|C \cap B_R(w)|}{|B_R(w)|}, \qquad d_{4.6.12} = \inf_C d(C),$$

where C ranges over all identifying codes in G. Sampaio obtained the lower bound 3/7 by discharging and the upper bound 13/30 from a periodic code with 52 code vertices among 120 vertices per fundamental region. [1, Theorems 6 and 10 and Figure 10]

Theorem 1 (Plane bounds). For every radius-one identifying code C in G and every $w \in V(G)$,

$$\liminf_{R \to \infty} \frac{|C \cap B_R(w)|}{|B_R(w)|} \geq \frac{1829}{4263}.$$

There exists a periodic identifying code with 135 code vertices among 312 vertices per fundamental region, of density 45/104. Consequently,

$$\frac{1829}{4263} \leq d_{4.6.12} \leq \frac{45}{104}.$$

Let $\mathrm{chg}_C(v)$ denote the final charge at v under Sampaio's rule. The rule guarantees $\mathrm{chg}_C(v) \geq 3$ at each vertex. Our additional estimate is that for every identifying code C and every center w,

$$\sum_{v \in B_{22}(w)} (\mathrm{chg}_C(v) - 3) \geq 2.$$

Section 3 sums this estimate over centers in a finite ball and passes to the lower limit. At each center the limsup is at least the liminf, so the lower bound survives both infima in the definition of $d_{4.6.12}$.

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Sections 4 and 5 prove that 45/104 is the exact minimum in two restricted classes. The first consists of codes invariant under the translation $(2, 1)$ in the honeycomb coordinates of Section 2. In the second, a bi-infinite sequence of anchor columns has successive gaps of two or three columns. Each anchor carries one of the 26 prescribed column patterns. The intervening one or two columns are subject only to the identifying-code conditions. Whether an unrestricted plane identifying code can have density below 45/104 remains open.

2. A periodic code with 135 marks

The kagome lattice is the line graph of the honeycomb lattice. Truncating each kagome vertex replaces it by a square and doubles the lengths of the triangular and hexagonal faces, giving the 4.6.12 grid. Index the black and white honeycomb vertices by A_q, B_q, with $q = (i, j) \in \mathbb{Z}^2$. The honeycomb edge of type $t \in \{0, 1, 2\}$ joins A_q to $B_{q+\delta_t}$, where $\delta_0 = (0, 0)$, $\delta_1 = (-1, 0)$ and $\delta_2 = (0, -1)$.

The square for this edge has four ports (q, t, r), numbered in cyclic order A^+, A^-, B^+, B^- by $r = 0, 1, 2, 3$. The sign indicates the next or previous edge type at that honeycomb endpoint. A port has two neighbors on its square, at $r \pm 1 \pmod 4$. For its third neighbor put $\varepsilon = (-1)^r$, $s = t + \varepsilon \pmod 3$ and $r' = r + \varepsilon$. This neighbor is (q, s, r') for $r < 2$ and $(q + \delta_t - \delta_s, s, r')$ for $r \geq 2$. In the latter case the two squares share the white endpoint $B_{q+\delta_t}$. Edge types are taken modulo three. Reversing the third edge changes the sign of ε and restores the starting port. Each port therefore has degree three.

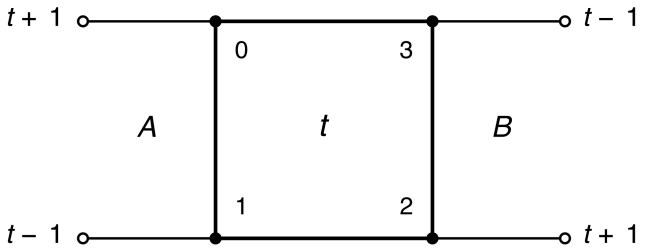


Figure 1. Filled dots belong to the central square, and open dots to its neighboring squares. The left and right bridges share the honeycomb endpoints A and B, respectively. Labels $t \pm 1$ give the type of the neighboring square. The diagram specifies adjacency only.

Index the following 26 words from zero, reading across the first line and then the second. At (q, t, r) take word $(2j - i) \bmod 26$ and read its t-th hexadecimal digit from the left, starting at zero. Select the vertex precisely when bit r of that digit is one, with bit zero the least significant bit.

```
2a9  2c9  8a7  8cd  478  926  e94  432  311  516  2a9  8cd  86d
96c  998  478  b84  c16  6c1  ad8  14b  cd8  165  b84  692  131
```

The phase map $(i, j) \mapsto 2j - i \pmod{26}$ is onto. Its kernel is generated by $(2, 1)$ and $(-26, 0)$, so $q = (i, 0)$, $0 \leq i < 26$, gives cell representatives. Each cell has twelve ports. The words contain 135 set bits in total, giving 135 marks among 312 vertices. The translates of this finite fundamental domain partition the graph. A domain that meets both a ball and its complement lies in a boundary shell of fixed thickness. The rank-two periodic graph has balls of size $\Theta(R^2)$ and spheres of size $O(R)$, uniformly in the center by vertex transitivity. In particular, $|B_{R+1}(w)|/|B_R(w)| \to 1$, the slow growth property of Sampaio, Sobral and Wakabayashi [3, Definition 3.1]. Thus

$$|C \cap B_R(w)| = \frac{135}{312}|B_R(w)| + O(R),$$

and the ball density at every center tends to $135/312 = 45/104$. [2, Corollary A.4]

For each of the 312 representative centers v, form the signature $I(v) = N[v] \cap C$ as a set of vertices in the infinite grid. Only the membership test uses the phase reduction. The numbers of representatives with signature sizes one, two and three are $129, 138, 45$, respectively. For each representative compare $I(v)$ with $I(u)$ for every $u \in \left(\bigcup_{z \in N[v]} N[z]\right) \setminus \{v\}$. All 2,496 directed comparisons give unequal signatures, including those across a period boundary. For any pair at distance at most two, a single period translation brings the first center into the representative set. Apply that same translation to the second center and to every vertex of both signatures, retaining their full coordinates. Centers farther apart have disjoint closed neighborhoods and both signatures are nonempty. The periodic set is therefore an identifying code on the entire plane. `portable_witness.py`

3. Forced surplus in a ball

Under Sampaio's rule each code vertex starts with charge 7 and sends nonnegative amounts to its neighbors in a single round. Every noncode vertex ends with charge 3. A code vertex receives at most one unit and ends with charge at least 3, so it sends at most 5 of its initial units. Put $e(v) = \mathrm{chg}_C(v) - 3$. The transfers are multiples of $1/2$, so $2e(v)$ is a nonnegative integer. [1, proof of Theorem 6, rules R1 and R2]

Lemma (Radius-22 surplus). For every identifying code C and every vertex w,

$$\sum_{v \in B_{22}(w)} e(v) \geq 2.$$

Finite proof. Vertex transitivity and invariance of the square and bridge rules allow us to fix one center w. A breadth-first enumeration gives $|B_{22}(w)| = 609$. Suppose the conclusion fails. Then $\sum_{B_{22}(w)} 2e(v) \leq 3$. We show that the code would give a satisfying assignment of the supplied finite Boolean formula.

Give the site variables their actual membership bits. The formula includes domination clauses for centers in $B_{24}(w)$ and separation clauses comparing such centers with vertices at distance one or two. The clauses read their full supports in the original graph, including sites outside $B_{24}(w)$. The code satisfies each clause by definition.

For each $v \in B_{22}(w)$, assign its eight markers by

$$M_{v,\ell} = 1 \quad \Longleftrightarrow \quad 2e(v) \geq \ell, \qquad 1 \leq \ell \leq 8.$$

Nine site bits determine the charge. They are the four vertices of the square of v, its bridge neighbor, the two square neighbors of that bridge neighbor, and the bridge neighbors of the two square neighbors of v. On all 2^9 bit patterns the verifier checks that whenever the site literals of a marker clause force $M_{v,\ell}$ to be true, the computed doubled surplus is at least ℓ. Thus the actual site bits and the assignment above satisfy every marker clause. The biconditional defines our choice of marker values when extending a plane code. Under the contradiction hypothesis every $2e(v)$ is a nonnegative integer at most three. The eight levels therefore record the full sum.

$$\sum_{v \in B_{22}(w)} \sum_{\ell=1}^{8} M_{v,\ell} = \sum_{v \in B_{22}(w)} 2e(v) \leq 3.$$

For each auxiliary counter bit Y the verifier reconstructs a marker set A_Y and an integer threshold q_Y. Assign

$$Y = 1 \quad \Longleftrightarrow \quad \sum_{M \in A_Y} M \geq q_Y.$$

For a counter implication, the verifier checks that the output support contains each input support and that the output threshold is at most the sum of the input thresholds. When two counts are added, it also checks that their input supports are disjoint. Hence true inputs make the output true under this assignment.

Each output forbidden by a negative unit clause has threshold greater than three and is false when the total marker weight is at most three. This supplies a satisfying extension for every plane code obeying the contradiction hypothesis.

For this formula of 20,393 variables and 125,538 clauses, the supplied DRUP trace verifies 83,781 clause additions and ends in the empty clause. The formula is unsatisfiable, contradicting the assignment above. □

From the lemma to the plane. Fix a center o and write $B_s = B_s(o)$. Sum the lemma over $w \in B_R$. A contributing vertex belongs to B_{R+22} and is counted for at most 609 centers, since $v \in B_{22}(w)$ if and only if $w \in B_{22}(v)$. Nonnegativity of e gives

$$2|B_R| \le 609 \sum_{v \in B_{R+22}} e(v).$$

Only code vertices supply initial charge, seven units each. Transfers are nonnegative and cross at most one edge in a single round, and each code vertex sends at most its initial charge. Every unit that ends in B_{R+22} therefore originated in B_{R+23}. Combining the charge balance with the surplus estimate gives

$$\begin{aligned} 7|C \cap B_{R+23}| &\ge \sum_{v \in B_{R+22}} \mathrm{chg}_C(v) \\ &= 3|B_{R+22}| + \sum_{v \in B_{R+22}} e(v) \\ &\ge 3|B_{R+22}| + \frac{2}{609}|B_R|. \end{aligned}$$

Divide by $7|B_{R+23}|$. For $j = 0, 22$, slow growth gives $|B_{R+j}|/|B_{R+23}| \to 1$ by [3, Lemma 3.2(a)]. This is the boundary estimate used for infinite-graph discharging in their Theorem 3.3. As $R + 23$ runs through all sufficiently large integer radii, the liminf at o is at least $(3+2/609)/7 = 1829/4263$. This proves the lower bound of Theorem 1 for every center without assuming a density limit.

4. A prescribed translation period

Theorem 2. Among identifying codes invariant under the translation $(2, 1)$ in honeycomb coordinates, the minimum density is $45/104$. No period is assumed in a second direction.

Proof. The integer $z = 2j - i$ labels the orbits of honeycomb cells under $(2, 1)$. Each orbit contains twelve types of ports, indexed by (t, r). For each domination or separation condition, first form its support in the original plane graph. In particular, take the symmetric difference of the two closed neighborhoods before identifying any vertices. Then replace each site in the support by its orbit bit. If several sites project to the same bit, their repeated positive literal can be merged without changing the clause.

The verifier reads twelve bits per layer. For port type (t, r) its scan layer is $\lambda = z + \sigma_{t,r}$, with a fixed integer offset and a fixed within-layer order supplied in the certificate. The checker reconstructs the plane clauses and verifies their indexed supports against the stored rules. At a layer boundary the state consists of the inclusion-minimal sets of unread positions of pending clauses. A selected bit satisfies each clause containing it. A transition is rejected if an unsatisfied clause runs out of positions. Starting from the empty state omits obligations introduced before the cut and can only enlarge the set of finite paths. Thus each finite block of an invariant plane code gives a legal path.

Let A be the certified set of 16,597 boundary states. The verifier checks that it contains the empty state and that all legal one-layer transitions from A have endpoint set A. For a path $p : s \to t$ through one layer, let $b(p)$ count its selected bits. The certificate gives integer functions $H_0, \dots, H_{25}$ on A. For each $t \in A$ the verifier checks

$$\min_{p : s \to t} \bigl(H_h(s) + b(p)\bigr) = \begin{cases} H_{h+1}(t), & 0 \le h < 25, \\ H_0(t) + 135, & h = 25. \end{cases}$$

Take the certificate phase at the cut before layer λ to be $h = \lambda \bmod 26$, and denote the state at that cut by s_λ. Fix an integer $m \geq 1$ and partition the layers into blocks

$$I_a = \{\lambda \in \mathbb{Z} : 26ma \leq \lambda < 26m(a+1)\}, \qquad a \in \mathbb{Z}.$$

Write $u = 26ma$ and $v = 26m(a+1)$. The cuts before layers u and v both have phase zero. The last layer in I_a has phase 25. The minimum-cost identity bounds every legal transition from below. Summing along a path across I_a cancels the intermediate potentials and adds 135 exactly m times. Its selected count is at least

$$135m + H_0(s_v) - H_0(s_u) \geq 135m - 6,$$

since $0 \leq H_0 \leq 6$. The endpoint states may differ, and the code bits need not repeat after 26 layers. `check_cylinder.py`

For every fixed j, t, r, layer λ corresponds to the unique plane vertex (i, j, t, r) with $i = 2j + \sigma_{t,r} - \lambda$. Conversely that vertex determines $\lambda = 2j - i + \sigma_{t,r}$. A tile consists of one fixed j, one block I_a and all twelve port types. These tiles partition the plane and each contains $12 \cdot 26m = 312m$ vertices. By $(2, 1)$-invariance, reading a tile gives the orbit bits on I_a, so each tile contains at least $135m - 6$ code vertices.

For fixed m these tiles are translates of a finite set. The tiles cut by the boundary of $B_R(w)$ occupy a shell of fixed thickness depending on m, of size $O_m(R)$ by Section 2. Keeping only whole tiles and taking the ball liminf at each center gives a lower bound of $(135m - 6)/(312m)$. Letting $m \to \infty$ gives $45/104$. The code in Section 2 is invariant under $(2, 1)$ and has this density. □

5. Arbitrary fillings between prescribed columns

A column consists of all cells with a fixed honeycomb coordinate i. Write C_* for the periodic code of Section 2. Phase $a \in \mathbb{Z}/26\mathbb{Z}$ prescribes the entire column word $P_a(j, t, r) = \mathbf{1}_{C_*}(a, j, t, r)$. Let i_p, $p \in \mathbb{Z}$, be a strictly increasing sequence of anchor columns with successive differences two or three. At i_p prescribe one of the words P_{a_p}. The intervening columns may have any filling that gives an identifying code on the plane.

Theorem 3. The minimum density in this family is $45/104$, with no periodicity assumption on the sequence of anchors or on the filling.

One slab. Let $w = i_{p+1} - i_p - 1 \in \{1, 2\}$ be the number of free columns and put $a = a_p$, $b = a_{p+1}$. Impose the two anchor words, but count vertices in the half-open interval $[i_p, i_{p+1})$. This includes the left anchor and all free columns. Each plane clause of Section 4 lies in one column or two consecutive columns, as can be checked on the twelve port types in one cell. The checker substitutes the anchor bits into the clauses supported on the closed slab. Every genuine plane code satisfies the resulting interior constraints.

The boundary words repeat after thirteen rows, and $P_{a+2}(j, t, r) = P_a(j-1, t, r)$. Translating both words vertically therefore reduces the 26^2 ordered phase pairs to the 52 pairs with $a \in \{0, 1\}$ and arbitrary b. Let $c(a)$ count the marks in thirteen rows of one anchor. The table in Section 2 gives $c(a) = 67$ for even a and 68 for odd a. Put $u = 135/2$ and define $\phi(0) = 0$ and $\phi(a+1) - \phi(a) = c(a) - u$. Thus ϕ alternates between 0 and $-1/2$ and is unchanged by the simultaneous even phase shift.

For each representative pair the verifier reconstructs the clauses on the free columns. One complete certificate transition scans 13 consecutive rows of those w columns, or $13 \cdot 12w = 156w$ site bits. Let d be the number of potential phases, K the cost increment after d transitions, and $\mu_w(a, b) = K/d$ the certified interior cost in marks per thirteen-row block. Write $\Psi_h(s) = H_h(s) - h\mu_w(a, b)$ for the adjusted potential of state s at phase h. The min-plus relation of Section 4 gives a cost of at least $\mu_w(a, b)$ plus the change in Ψ on each complete transition, including the wrap from phase $d - 1$ to zero.

Let M_{free} count the marks in the free columns on any legal path through N consecutive complete blocks aligned in the representative frame. Summing the transition inequalities gives

$$M_{\text{free}} \geq N\mu_w(a,b) + \Psi_{\text{out}} - \Psi_{\text{in}} \geq N\mu_w(a,b) - D_w.$$

Here D_w bounds the range of Ψ over all states and phases for each width-w certificate. The largest such ranges are $D_1 = 2$ and $D_2 = 4$. The initial cut can use the empty state. The filling may vary from block to block. `check_slabs.py` and `aggregate.py`.

In these same $13N$ rows the left anchor contributes exactly $Nc(a)$ marks. The right anchor enters the constraints and is counted in the next slab. Define $s_w(a,b) = c(a) + \mu_w(a,b) - (w+1)u - \phi(b) + \phi(a)$. The 52 representative checks for each width give $s_w(a,b) \geq 0$, with equality precisely for $b = a + w + 1 \pmod{26}$. The cost of the half-open slab satisfies

$$\begin{aligned} M_{\text{slab}} = Nc(a) + M_{\text{free}} &\geq N\big(c(a) + \mu_w(a,b)\big) - D_w \\ &= N\big((w+1)u + \phi(b) - \phi(a) + s_w(a,b)\big) - D_w. \end{aligned}$$

The choice of initial row for this alignment can depend on the pair. For example, when $w = 2$ and $(a,b) = (0,3)$, the left anchor contributes 67 and the interior rate is 135. Their sum 202 is half a unit below $3u$ and exactly equals $3u + \phi(3) - \phi(0)$.

Joining slabs. Consider a rectangle of W columns and H rows, with $W, H \geq 26$. Keep only whole half-open slabs inside it, losing at most two columns at each side. In each retained slab take exactly $N = \lfloor H/13 \rfloor - 1$ consecutive aligned blocks that fit within the rectangle. Such a choice is possible for every alignment, and $H - 13N \leq 25$. Index the retained slabs by $i = 0, \ldots, s-1$. Write $q_i = w_i + 1$ for their half-open widths and $a_0, \ldots, a_s$ for the original anchor phases. A common even shift of a pair of phases leaves its ϕ difference unchanged. Thus, even when the chosen blocks start at different rows, each shared phase has the same coefficient N on both sides. Summing the M_i for the retained portions and dropping the nonnegative s_{w_i} gives

$$|C \cap \text{rectangle}| \geq \sum_{i=0}^{s-1} M_i \geq Nu \sum_{i=0}^{s-1} q_i + N\big(\phi(a_s) - \phi(a_0)\big) - \sum_{i=0}^{s-1} D_{w_i}.$$

We have $\sum_i q_i \geq W - 4$, $|\phi(a_s) - \phi(a_0)| \leq 1/2$, $\sum_i D_{w_i} \leq 4s \leq 4W$ and $13N \geq H - 25$. Hence the rectangle contains at least

$$\frac{u}{13}WH - O(W+H) = \frac{45}{104}\,12WH - O(W+H)$$

marks, uniformly in its origin, the phases and the filling.

For fixed $M \geq 26$, tile the plane by $M \times M$ cell squares. The whole tiles inside a graph ball have density at least $45/104 - O(1/M)$. The tiles meeting its boundary occupy $O_M(R)$ vertices. Taking first the ball liminf as $R \to \infty$ and then $M \to \infty$ gives the lower bound at every center. The code C_* itself belongs to the family, for instance with anchors $i_p = 2p$ carrying their original column words. It attains $45/104$. □

6. Reproducibility and the unrestricted problem

The certificate archive contains the finite data and checkers for Sections 2–5. The checks have distinct mathematical roles.

Section	What the reader can verify
2. Plane upper bound	Decode the 26 words and count 135 selected vertices among 312 representatives. Compare closed-neighborhood signatures in the infinite graph. This check uses only the port rules and the words printed in Section 2.
3. Plane lower bound	`check_original.py` checks the geometric clauses, marker implications and counter thresholds of the finite formula. For that same formula, `check_drup_stream.py` verifies the DRUP trace through the empty clause.
4. One translation period	`check_cylinder.py` regenerates the projected plane clauses and verifies the transition inequalities. For the translation $(2, 1)$, the potential has 26 phases and a cost increment of 135. The code may vary without a period in the transverse direction.
5. Prescribed columns	`check_slabs.py` verifies the costs for 52 representative pairs of anchor phases at each free width. These representatives come from simultaneous translations of the two anchor words. The fillings and the sequence of slabs may be nonperiodic. `aggregate.py` computes the endpoint losses from the potentials.

The checkers establish the finite premises in the table. Section 3 uses the first lower-bound check to extend any plane code with insufficient surplus to a satisfying assignment of the formula. The second check excludes that assignment. The counting and limiting arguments in Sections 3–5 carry the resulting finite estimates to arbitrary codes in the respective classes.

The replay instructions in `VERIFY.md` use Python 3.12, with NumPy for the transition checks and the MiniSat22 backend of `python-sat` for unit propagation in the DRUP replay. The archive is `IC_4612_DENSITY_CERTIFICATES_20261102.zip`.

Two surplus values account for the endpoints of our interval. Over one fundamental region of the periodic code C_*, transfers across opposite boundaries cancel. Its mean surplus is therefore

$$\frac{7 \cdot 135 - 3 \cdot 312}{312} = \frac{3}{104}.$$

For every identifying code C and every center w, the averaging argument of Section 3 gives the universal estimate

$$\liminf_{R \to \infty} \frac{1}{|B_R(w)|} \sum_{v \in B_R(w)} e(v) \geq \frac{2}{609}.$$

Sections 4 and 5 prove that every code in their respective restricted classes has density at least $45/104$. The periodic code C_* belongs to both classes and attains equality. On the unrestricted plane we have

$$\frac{1829}{4263} \leq d_{4.6.12} \leq \frac{45}{104}.$$

The open question is whether $d_{4.6.12} < 45/104$, equivalently whether there exists a plane identifying code C with $d(C) < 45/104$.

References

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